

Some review questions - 12th week

Q1 Give the definitions of eigenvalues and eigenvectors.

Q2 Give the definition of the characteristic polynomial of a matrix A .

Q3 Prove that the eigenvalues of a matrix are the roots of its characteristic polynomial.

Q4 Give the different steps required to compute the eigenvalues and eigenspaces of a matrix $A \in \mathbb{R}^{n \times n}$.

Q5 Provide the eigenvalues and associated eigenspaces of $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$.

Q6 Provide the eigenvalues of $A = \begin{bmatrix} 2 & 1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{bmatrix}$.

Q7 Provide the eigenvalues and associated eigenspaces of $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Q8 Provide the eigenvalues of $A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix}$.

Q9 State the Fundamental Theorem of Algebra (about the roots of polynomials) and explain its consequences for the number of eigenvalues of a matrix.

Q10 Explain the difference between a set and a multiset. Why is this important for eigenvalues?

Q11 Given a complex number $z = a + bi$, where $a \in \mathbb{R}$ is the real part and $b \in \mathbb{R}$ is the imaginary part, (i) give a characterization of the number denoted by “ i ”, (ii) give the definition of the complex conjugate $\bar{z}$, and (iii) express the modulus of z , first in terms of $a, b \in \mathbb{R}$, and then in terms of z and $\bar{z}$.

Q12 Is the standard basis of $\mathbb{R}^3$, $\{e_1, e_2, e_3\}$, also a basis of $\mathbb{C}^3$?

Q13 Are the vectors $u = \begin{bmatrix} 1 \\ \mathbf{i} \end{bmatrix}$ and $v = \begin{bmatrix} \mathbf{i} \\ -1 \end{bmatrix}$ linearly independent? Prove your result.

Q14 Give the definition of an eigenbasis.

Q15 Give the eigenvalue(s) and associated eigenspace(s) of $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Does A admit an eigenbasis?

Q16 Give the eigenvalue(s) and associated eigenspace(s) of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Does A admit an eigenbasis?

Q17 Given the diagonal matrix $D = \begin{bmatrix} \lambda_1 & & (0) \\ & \ddots & \\ (0) & & \lambda_n \end{bmatrix}$ and a matrix $X \in \mathbb{C}^{n \times n}$, state a simple rule to express DX and XD .

Q18 Consider a matrix $A \in \mathbb{C}^{n \times n}$ admitting n eigenvectors $v_1, \dots, v_n \in \mathbb{C}^n$ associated with the eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{C}$. Express in matrix form the identities $Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n$.

Q19 Show that if a matrix $A \in \mathbb{R}^{n \times n}$ admits an orthonormal eigenbasis of $\mathbb{R}^n$, then A is symmetric.

Q20 Give the precise statement of the spectral theorem for symmetric matrices.

Q21 Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$, show that two eigenvectors of A corresponding to two different eigenvalues are orthogonal.